# [12] What Can Second-Order Information Buy?

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Second-Order Information in Non-Convex Stochastic Optimization: Power and Limitations — RQ3, RQ4

**Problem:** How many stochastic oracle calls are *unavoidably* necessary for finding approximate stationary points? Does access to Hessian information fundamentally help, and can even higher derivatives do better? **Method:** Matching algorithms and worst-case lower-bound constructions in stochastic optimization. **Result:** An $O(\epsilon^{-3})$ optimal stochastic gradient/Hessian-vector rate for first-order stationarity; nearly matching bounds also for second-order targets.

## 1. Why study oracle complexity?

In optimization, an *oracle* answers requests about a function. A stochastic gradient oracle returns $g(x;\xi)$; a stochastic Hessian-vector oracle returns a noisy version of $\nabla^2 F(x)v$ without forming the full Hessian matrix. Counting calls abstracts away software and hardware differences and asks what information an algorithm *must* acquire.

$$\text{Goal (FOSP):} \quad \|\nabla F(\hat{x})\| \le \epsilon.$$

This criterion is weaker than an SOSP, because negative curvature can remain.

## 2. From a noisy gradient to the value of curvature

For twice differentiable $F$, Taylor's theorem says

$$\nabla F(x+\delta v) = \nabla F(x) + \delta \nabla^2 F(x)v + O(\delta^2)$$

when the Hessian is sufficiently regular. An HVP reveals how gradients change in direction $v$. Repeated HVPs can detect a direction of negative curvature, but in this paper they also help produce *more accurate first-order progress* using stochastic second-order structure.

## 3. The principal matching result

The authors construct a stochastic second-order method using stochastic gradients and Hessian-vector products that finds an $\epsilon$-FOSP with

$$\boxed{O(\epsilon^{-3}) \text{ stochastic derivative-oracle queries}}$$

up to fixed smoothness/noise/problem constants in the stated oracle model. They prove a matching lower bound for that model. Remarkably, access to derivatives of any fixed order $p \ge 2$ does not improve the worst-case exponent below 3, even with suitable Lipschitz regularity of the first $p$ derivatives.

**Meaning of a lower bound.** An *upper bound* constructs an algorithm needing at most a stated number of calls; a *lower bound* constructs hard functions on which every permitted algorithm needs at least that order. When upper and lower exponents match, the rate is optimal *for that oracle/assumption class*. This is not a statement about every training dataset.

## 4. The second-order target is a separate result

They additionally consider

$$\|\nabla F(\hat{x})\| \le \epsilon, \qquad \lambda_{\min}(\nabla^2 F(\hat{x})) \ge -\gamma.$$

The paper establishes nearly matching upper/lower bounds for this $(\epsilon,\gamma)$-SOSP task. **Crucial distinction:** The headline $O(\epsilon^{-3})$ theorem is about *first-order stationarity*. It would be incorrect to quote that rate without qualification as a universal SOSP guarantee, or as a rate for finding a global minimum.

## 5. Assumptions and boundaries

These are general smooth stochastic objectives with specified stochastic derivative access and regularity/noise controls. Their lower-bound constructions establish worst-case hardness for the allowed algorithm family, not predictions for actual finite neural networks trained on modern accelerators. Hessian-vector evaluations may have different computational costs than ordinary gradient evaluations; memory, parallelism and tuning are not represented by a single oracle count.

**Survey caution.** Never compare $O(\epsilon^{-3})$ from this paper with $\widetilde{O}(\epsilon^{-7/4})$ for deterministic accelerated escape without stating the different *stochastic versus deterministic oracle* and *FOSP versus SOSP* targets.

**Our takeaway.** Curvature information provably improves what can be achieved with noisy optimization, but there is a fundamental limit: arbitrarily higher derivatives do not automatically yield arbitrarily faster convergence.
**Source:** https://proceedings.mlr.press/v125/arjevani20a.html.